Day 3 Report: Core Feature Implementation

1. Title

Mental Health First-Response Chatbot – Day 3 Progress Report
Focus: Sentiment Detection, Crisis Escalation, and Tier-Based Response System

2. Objective

The objective of Day 3 was to enhance the chatbot by adding intelligence and safety features. This includes detecting user emotions, identifying crisis situations, and generating appropriate responses based on severity levels.

3. Overview of Work Done

On Day 3, we implemented a tier-based response system that classifies user input into three levels: normal, emotional, and critical. The system first checks for crisis-related keywords to ensure user safety. If no crisis is detected, it analyzes the emotional state of the user and provides responses accordingly using either predefined chatbot responses or CBT-based advice through the RAG system developed in Day 2.

4. Key Components Implemented

- **Crisis Detection**

- Detects high-risk inputs such as "I want to die", "suicide", etc.
- Uses keyword-based matching
- Immediately triggers an emergency response

- **Sentiment Detection**

- Identifies user emotions such as:
  - Sad
  - Anxious
  - Neutral
- Helps in understanding user mental state

- **Tier-Based System**

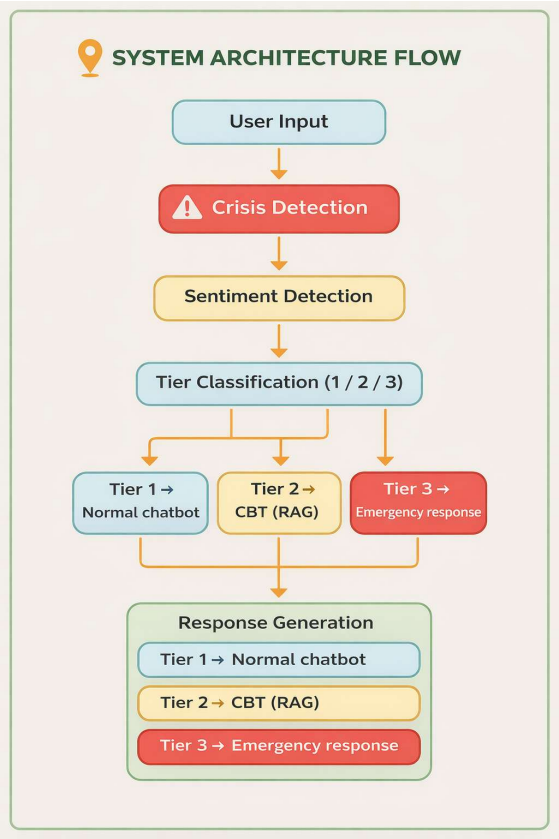
The chatbot classifies input into three tiers:

Tier	Description	Response Type
Tier 1	Normal Input	Basic chatbot response
Tier 2	Emotional Distress	CBT-based advice (RAG)
Tier 3	Crisis Situation	Emergency helpline message

Final Chatbot Logic

- Combines crisis detection, sentiment analysis, and RAG
- Ensures safe and relevant responses
- Prevents unsafe or misleading outputs

5. System Architecture Flow



6. Technologies Used

- Python
- LangChain
- ChromaDB (Vector Database)
- HuggingFace Embeddings
- Sentence Transformers
- RAG (Retrieval-Augmented Generation)

7. Sample Outputs

Input:
"hello"
Output:
Normal chatbot response

Input:
"I feel stressed"
Output:
CBT-based suggestion (RAG)

Input:
"I want to die"
Output:
Emergency helpline message

8. Achievements

- Successfully implemented crisis detection system
- Built a tier-based decision model
- Integrated Day 2 RAG system with intelligent logic
- Ensured user safety through emergency handling

9. Conclusion

The Day 3 implementation successfully transformed the chatbot into a safer and more intelligent system. By incorporating crisis detection and emotional understanding, the chatbot can now respond appropriately based on user needs, ensuring both usefulness and safety.

10. Future Scope

- Improve sentiment detection using advanced BERT models
- Expand crisis detection with more keywords and patterns
- Integrate real-time alerts (SMS/Email using APIs)
- Develop frontend interface for better user interaction